

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1063

Mount Jackson 138 kV
August 23, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1063. The Interconnection Customer proposes a 175.3 MW solar photovoltaic facility interconnecting at the Mount Jackson 138 kV substation in the Virginia area. The request was received on September 12, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 291 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$48,652,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LURAY HYDRO STATION - PURDY 138kV	CHATOM - PURDY 138kV	133.2	210	27.9
LURAY HYDRO STA - PUDDLEDOCK RD 138kV	CHATOM SUBSTATION - PURDY 138kV L	132.9	210	27.9
BROOKLYN CITY N PLANT - INTERNATIONAL PAPER VALLIANT OK 138kV	AMERICAN - URBAN 345kV	123.4	294	5.2
NICHOLS - ST LOUIS 138kV	PURDY - PAGE SUBSTATION 138kV	123.1	346	16.6
ROOSEVELT RD - ELKTON D.P 138kV line	ST LOUIS - NUCLEAR FUELS 138kV	121.2	251	9.9
ELKTON D.P - HOUSTON 138kV	ST LOUIS - NUCLEAR FUELS 138kV	121.2	251	9.9
WADSWORTH - ROOSEVELT RD 138kV	ST LOUIS SUBSTATION - NUCLEAR FUELS 138kV	119.9	260	9.9
S BELLEVILLE - NICHOLS 138kV	PAGE - PURDY 138kV	119.6	352	16.6
PUDDLEDOCK RD - MOUNTJACKSON SUBSTATION 138kV	PURDY - CHATOM 138kV	118.8	240	27.9
DETOR - SILOAM SPRINGS 138kV	FT BENTON - DETOUR 138kV	115.2	253	14.5
DETOUR - CEDARVILLE JCT 138kV	ST LOUIS - NICHOLS 138kV L	114.5	411	21.0
PAGE - CEDARVILLE JUNCTION 138kV line	ST LOUIS - NICHOLS 138kV line	114.2	416	21.0
PURDY - PAGE 138kV Line	ST LOUIS - NICHOLS 138kV L	113.3	431	21.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DETOUR SUBSTATION - FT BENTON SUBSTATION 138kV	DETOUR - SILOAM SPRINGS 138kV	112.3	315	15.7
VIDEO PD - MCAULEY 138kV line	ST LOUIS - NICHOLS 138kV	110.1	228	5.7
BROOKLYN CITY NORTH PLANT - MIDSET 138kV	AMERICAN - URBAN 345kV line	107.5	342	5.2
ABBOTSFORD - BRIDGE RIV 138kV	AMERICAN - URBAN 345kV Line	107.5	312	8.1
MOUNT PLEASANT - S JORDAN 138kV	MOUNT PLEASANT - FIREBANTZ 138kV line	104.6	188	6.2
SPRINGFIELD POWER LLC - EDINBURG 138kV Line	AMERICAN - URBAN 345kV	104.3	175	16.8
SILOAM SPRINGS - LEWISBURG 138kV Line	DETOUR - FT BENTON 138kV	100.7	307	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PURDY - LURAY HYDRO STA 138kV line	Rebuild line with higher-capacity conductor	\$2,497,000	79% (\$1,979,000)
LURAY HYDRO STA - PUDDLEDOK RD 138kV	Rebuild line with higher-capacity conductor	\$4,671,000	79% (\$3,701,000)
BROOKLYN CITY N PLANT - INTERNATIONAL PAPER VALLIANT OK 138kV	Rebuild line with higher-capacity conductor	\$1,736,000	22% (\$384,000)
ST LOUIS - NICHOLS 138kV	Rebuild line with higher-capacity conductor	\$3,096,000	19% (\$576,000)
ROOSEVELT RD - ELKTON D.P 138kV	Rebuild line with higher-capacity conductor	\$1,205,000	17% (\$201,000)
HOUSTON - ELKTON D.P 138kV L	Rebuild line with higher-capacity conductor	\$1,754,000	17% (\$292,000)
WADWORTH - ROOSEVELT RD 138kV	Rebuild line with higher-capacity conductor	\$4,227,000	17% (\$704,000)
S BELLEVILLE - NICHOLS 138kV	Rebuild line with higher-capacity conductor	\$6,593,000	19% (\$1,226,000)
PUDDLEDOK RD - MOUNTJACKSON SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$7,654,000	79% (\$6,065,000)
DETOR - SILOAM SPRINGS 138kV	Rebuild line with higher-capacity conductor	\$19,358,000	54% (\$10,355,000)
DETOUR - CEDARVILLE JCT 138kV	Rebuild line with higher-capacity conductor	\$10,940,000	41% (\$4,479,000)
PAGE - CEDARVILLE JUNCTION 138kV line	Rebuild line with higher-capacity conductor	\$8,989,000	41% (\$3,680,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PURDY - PAGE 138kV Line	Rebuild line with higher-capacity conductor	\$4,414,000	41% (\$1,807,000)
FT BENTON - DETOUR 138kV	Rebuild line with higher-capacity conductor	\$6,005,000	53% (\$3,203,000)
VIDEO PD - MCAULEY 138kV line	Rebuild line with higher-capacity conductor	\$2,376,000	35% (\$841,000)
BROOKLYN CITY NORTH PLANT - MIDSET 138kV	Reconductor line	\$3,761,000	22% (\$836,000)
ABBOTSFORD - BRIDGE RIV 138kV	Reconductor line	\$3,739,000	80% (\$2,992,000)
MOUNT PLEASANT - S JORAN 138kV L	Reconductor line	\$3,194,000	71% (\$2,267,000)
SPRINGFIELD POWER LLC - EDINBURG 138kV Line	Reconductor line	\$2,606,000	65% (\$1,682,000)
SILOAM SPRINGS - LEWISBURG 138kV L	Reconductor line	\$1,924,000	72% (\$1,382,000)
	Total allocated to project		\$48,652,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.